B

5. A process is currently running on the CPU. It then issues a disk I/O request and must wait until the I/O operation completes before it can continue. When the I/O operation completes, which state does the process usually enter first?
(A) Running
(B) Ready
(C) Waiting
(D) Terminated
(E) Suspended

C

6. An operating system requires that if a process needs multiple resources, it must request all of them before execution begins. If it cannot obtain all required resources at once, it will not hold any partial set of resources. Which necessary condition for deadlock does this design mainly break?
(A) Mutual exclusion
(B) No preemption
(C) Hold and wait
(D) Circular wait
(E) Starvation

B

7. An Ethernet frame transmitted on the wire begins with a 7-byte preamble (alternating bits 10101010...) followed by a 1-byte start frame delimiter. What is the primary purpose of the preamble?
(A) It allows the receiving adapter to synchronize its clock with the incoming bit stream before the actual frame content arrives
(B) It carries the destination MAC address so that switches can begin forwarding the frame earlier
(C) It provides error detection for the whole frame, similar to a checksum
(D) It encrypts the MAC addresses in the header to protect them from eavesdropping
(E) It pads the frame so that it reaches the minimum frame size of 64 bytes

p → NewNode → C ⇒ D　　　P　NewNode

A → B → C → D → Null

D

8. A linked list is currently:
A → B → D → NULL
Pointer p points to node B, and pointer newNode points to a new node C. Which instruction sequence correctly inserts C between B and D?
(A) p = newNode;　newNode → next = p → next
(B) p → next = newNode;　newNode → next = p → next
(C) newNode → next = p → next;　p → next = newNode
(D) newNode = p → next;　p → next = newNode
(E) p → next = p;　newNode → next = NULL